

Digital Control I: Sampling, Zero-Order Hold, and Discretization

Turning a continuous plant into a model a computer can control.

Ogata, *Modern Control Engineering*, chapters on digital control.

A digital controller sees the plant only at the sampling instants $t = kT$, and holds its command constant between samples through a **zero-order hold** (ZOH). Before we can design anything, we need a discrete-time model of the plant. In this tutorial you will:

- sample a continuous plant and build its discrete model with `c2d`,
- compare the **zero-order hold** and **Tustin (bilinear)** discretizations,
- see how continuous poles map to discrete poles through $z = e^{sT}$, and
- watch accuracy degrade as the sample time T grows.

Run with `publish('Intro.m')`, or step through with **Ctrl+Enter**.

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The continuous plant

We start from a lightly damped second-order plant

$$G(s) = \frac{1}{s^2 + 0.6s + 1}.$$

```
G = tf(1,[1 0.6 1]);
```

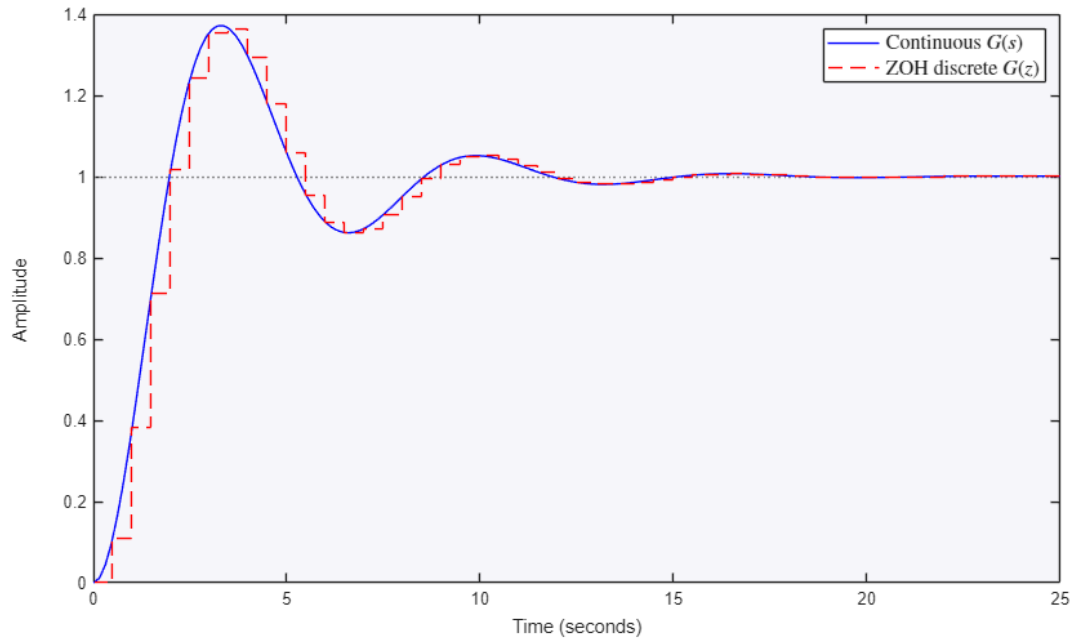
Sampling with a zero-order hold

The ZOH-equivalent discrete model `Gd` reproduces the continuous step response **exactly at the sampling instants**. Choose a sample time T short compared with the plant's natural period.

```
T = 0.5; % sample time (seconds)
Gd = c2d(G, T, 'zoh'); % zero-order-hold discretization

figure
step(G,'b', Gd,'r--')
legend('Continuous $G(s)$','ZOH discrete $G(z)$','Interpreter','latex','FontSize',12)
title(sprintf('Step Response: Continuous vs. ZOH ($T= %.2f$ s)',T), ...
'Interpreter','latex','FontSize',15)
```

Step Response: Continuous vs. ZOH ($T = 0.50$ s)



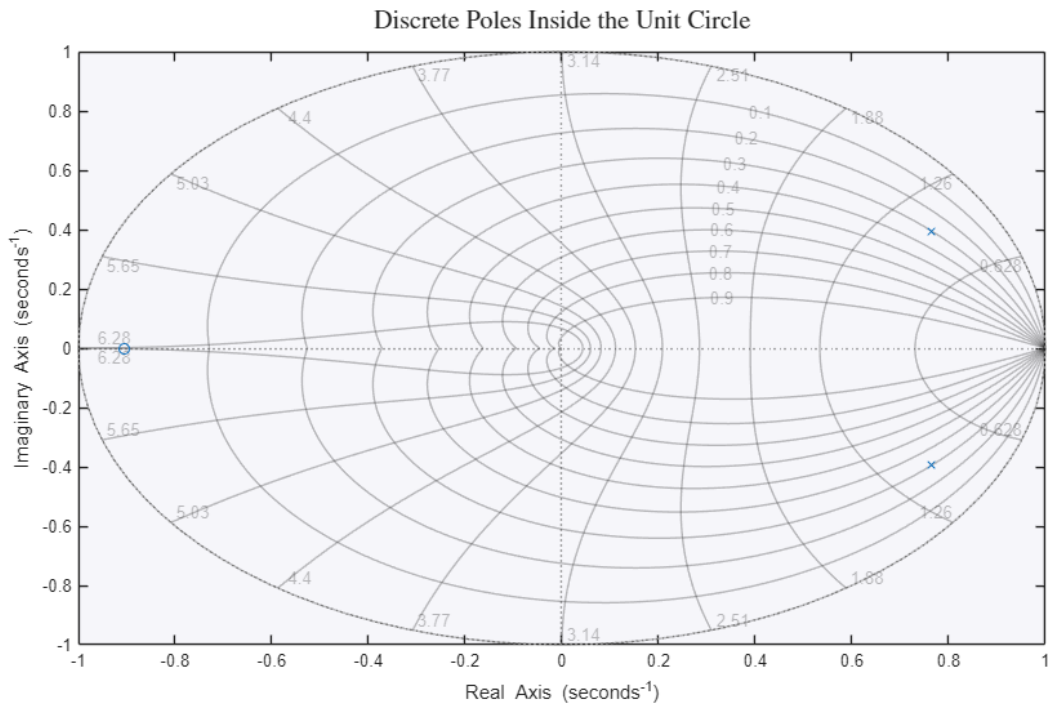
Where did the poles go? The $z = e^{sT}$ map

Sampling maps each continuous pole s_i to a discrete pole $z_i = e^{s_i T}$. Stable left-half-plane poles ($\text{Re } s < 0$) land **inside** the unit circle, which is the discrete-time stability region.

```
s_poles = pole(G);
z_mapped = exp(s_poles*T);
fprintf('Continuous poles: %s\n', mat2str(s_poles.',4))
fprintf('Mapped e^{sT}:    %s\n', mat2str(z_mapped.',4))
fprintf('c2d poles:        %s\n', mat2str(pole(Gd).',4))

figure
pzmap(Gd) % for a discrete model, pzmap draws the unit circle
title('Discrete Poles Inside the Unit Circle','Interpreter','latex','FontSize',15)
grid on

Continuous poles: [-0.3+0.9539i -0.3-0.9539i]
Mapped e^{sT}:    [0.7646+0.3951i 0.7646-0.3951i]
c2d poles:        [0.7646+0.3951i 0.7646-0.3951i]
```

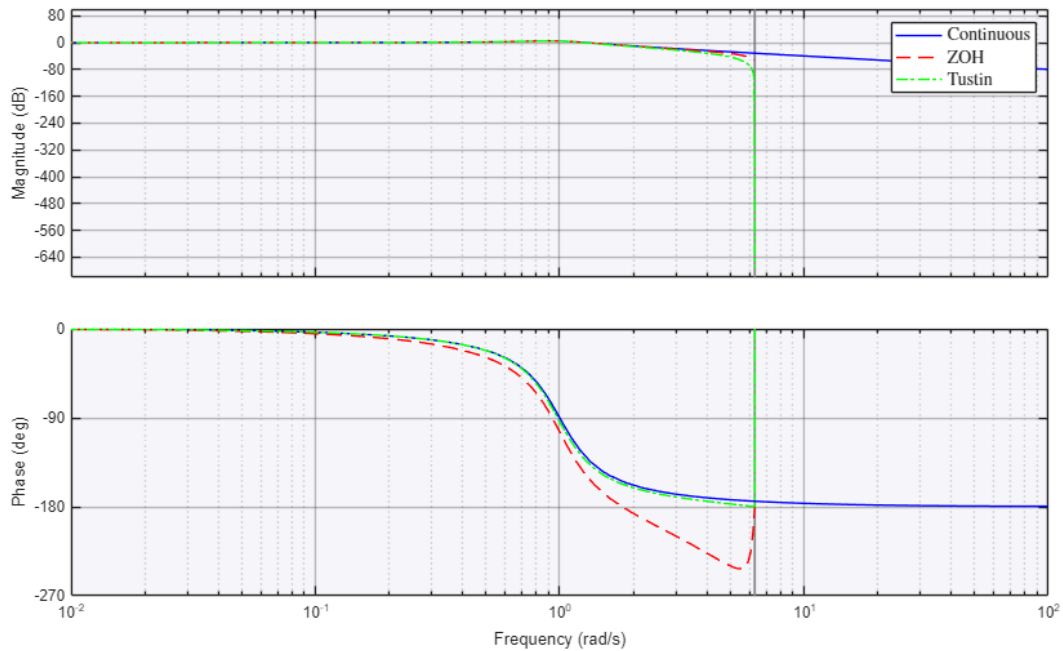


ZOH vs. Tustin (bilinear) discretization

`c2d` supports several methods. **ZOH** matches the step response at the samples; **Tustin** applies the bilinear rule $s \rightarrow \frac{2}{T} \frac{z-1}{z+1}$, which better preserves the frequency response and never maps a stable pole outside the unit circle.

```
Gd_tustin = c2d(G, T, 'tustin');
figure
bode(G,'b', Gd,'r--', Gd_tustin,'g-.')
legend('Continuous','ZOH','Tustin','Interpreter','latex','FontSize',12)
title('Frequency Response: Discretization Methods','Interpreter','latex','FontSize',15)
grid on
```

Frequency Response: Discretization Methods

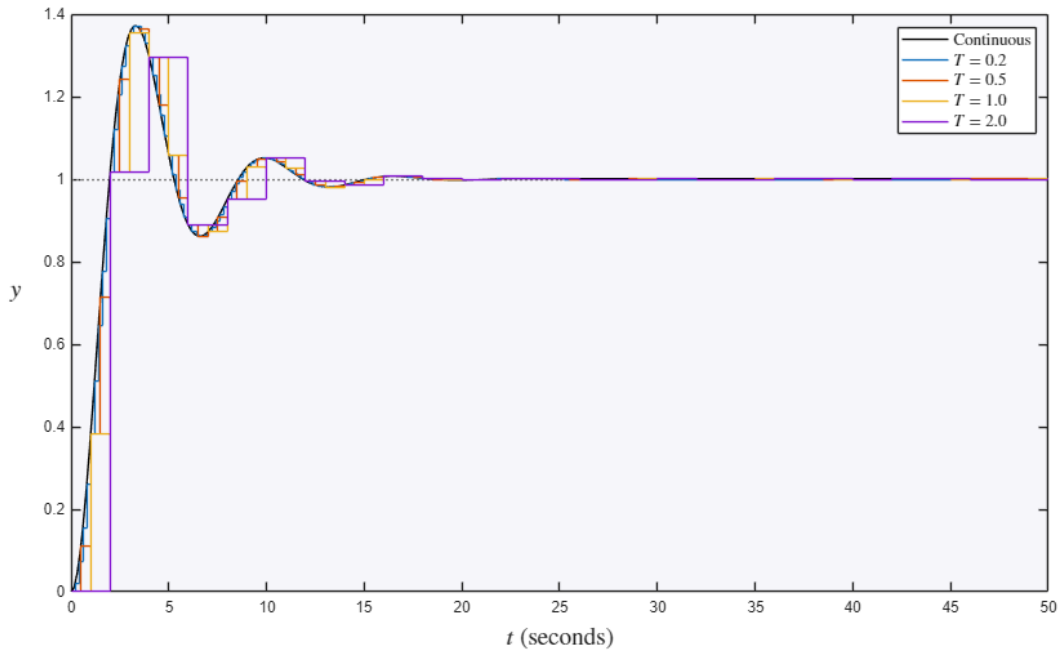


How fast must we sample?

A rule of thumb is to sample 20–30 times faster than the closed-loop bandwidth. Re-discretize at several sample times and overlay the step responses: as T grows, the discrete model – and any controller built from it – drifts away from the continuous truth.

```
figure
hold on
step(G,'k')
for Ti = [0.2 0.5 1.0 2.0]
    step(c2d(G,Ti,'zoh'))
end
hold off
legend('Continuous','$T=0.2$','$T=0.5$','$T=1.0$','$T=2.0$', ...
    'Interpreter','latex','FontSize',11)
title('Coarser Sampling Degrades the Discrete Model','Interpreter','latex','FontSize',15)
ylabel('$y$','Interpreter','latex','FontSize',16)
set(get(gca,'YLabel'),'Rotation',0)
xlabel('$t$','Interpreter','latex','FontSize',16)
```

Coarser Sampling Degrades the Discrete Model



Summary

- `c2d(G,T,method)` builds a discrete model: **ZOH** matches the sampled step, **Tustin** matches the frequency response.
- Continuous poles map to $z = e^{sT}$; the unit circle is the discrete stability boundary.
- Sample fast relative to the dynamics – coarse sampling throws away accuracy and, eventually, stability.

Next: `DiscretePID.m` runs a controller in discrete time, and `DeadbeatControl.m` exploits sampling to settle in a finite number of steps.

Try it yourself

- Halve the sample time `T` and notice the ZOH discrete step hug the continuous curve more tightly.
- Discretize with `'matched'` instead of `'zoh'` and compare the poles against `exp(s.poles*T)`.